

9(a)	time for activity (of sample) to halve	B1
9(b)(i)	activity (of X at time t)	B1
9(b)(ii)	- Y is a stable isotope - total number of nuclei is constant - half-life (of X) is 13.6 s - decay constant (of X) is 0.051 s^{-1} - amount (of X) at $t = 0$ is 0.066 mol - activity (of X) at $t = 0$ is $2.0 \times 10^{21} \text{ Bq}$ <i>Any three points, 1 mark each</i>	B3
9(c)	mass of 1 nucleus = $(7.3 \times 10^{-4}) / (4.0 \times 10^{22})$	C1
	nucleon number = mass of nucleus / (1.66×10^{-27})	C1
	$= (7.3 \times 10^{-4}) / (4.0 \times 10^{22} \times 1.66 \times 10^{-27})$ $= 11 \text{ and given as an integer}$	A1